Summary in Graph

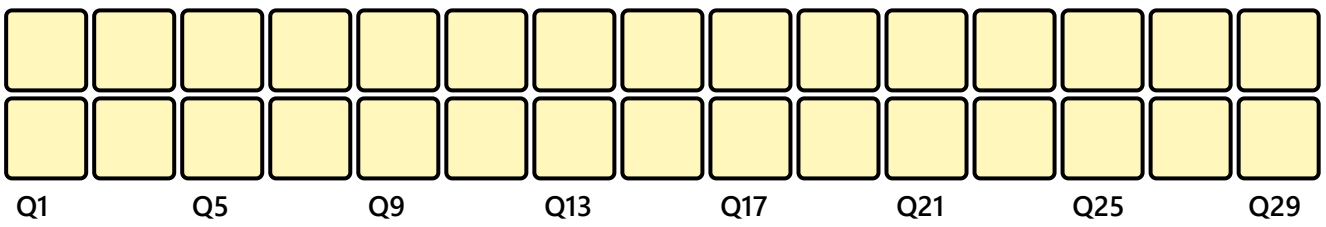
Exam Summary (GATE Overflow CS Test Series 2026 | Discrete Mathematics | Shallow Test 3)

Qs. Attempted:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)
Correct Marks:	0 (0 + 0)
Penalty Marks:	0 (0 + 0)
Resultant Marks:	0 (0 + 0)

Total Questions:	30 (10 + 20)
Total Marks:	50 (10 + 40)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (1) out of 11 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Numerical Type | Award: 1 | Penalty: 0 | Discrete Mathematics

Let b be an element in a group such that $\mathrm{ord}(b) = 3030$. The order of b^{57} is _____.

Your Answer: | Correct Answer: 1010:1010 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 Numerical Type | Award: 1 | Penalty: 0 | Discrete Mathematics

Let K be a group with order 72, and M a non-abelian subgroup of K. Assuming $M \neq K$, what is the smallest possible order of M?

Your Answer: | Correct Answer: 6:6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Discrete Mathematics

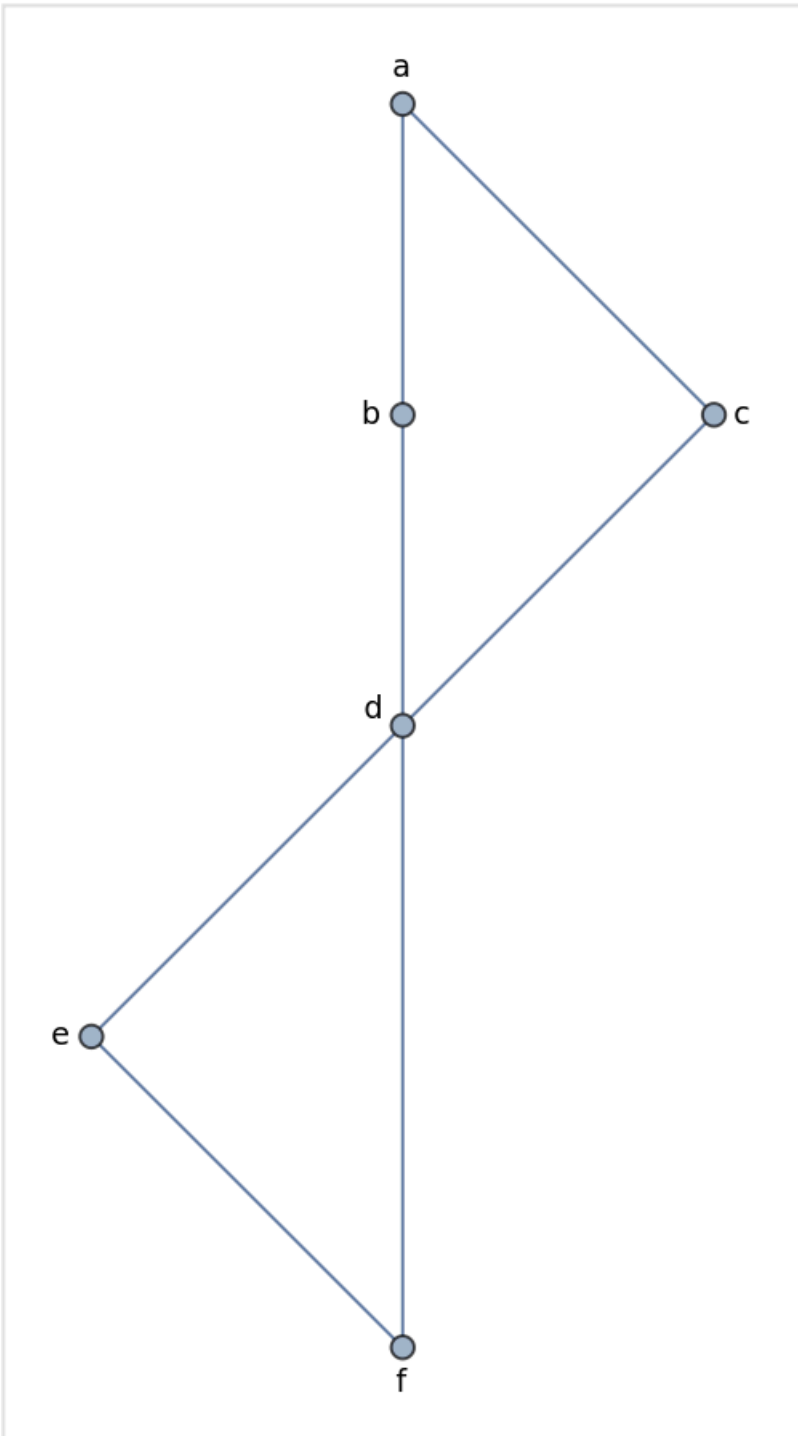
Let $G = \langle b \rangle$. The smallest subgroup of G containing b^{1440} and b^{1056} is _____

A. $\langle b^{36} \rangle$
B. $\langle b^{48} \rangle$
C. $\langle b^{72} \rangle$
D. $\langle b^{96} \rangle$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 Numerical Type | Award: 1 | Penalty: 0 | Mathematical Logic

Consider the below lattice.



The number of complements of element 'd' is____

Your Answer: | Correct Answer: 0:0 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Discrete Mathematics

Which of the following are correct? (Mark all the appropriate choices)

A. $(\mathbb{C}, +)$, the set of complex numbers under addition operation is a group.

B. $(\mathbb{N}, \cdot)$, the set of natural numbers under multiplication operation is a group.

C. $(\mathbb{Z}, \times)$, the set of integers under multiplication operation is a group.

D. $(\mathbb{Z}_p, \times)$, the set of integers $\{1, 2, \ldots, p-1\}$ where p is a prime number, under multiplication modulo p operation is a group.

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Mathematical Logic

What is the solution to the recurrence relation $c_n = c_{n-1} + 3n^2$ with $c_0 = 0$? A. $c_n = \frac{n(n+1)(2n+1)}{2}$ B. $c_n = n^2(n+1)$ C. $c_n = \frac{n(n+1)}{2}$ D. $c_n = n(n+1)(2n+1)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 02sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Discrete Mathematics

Consider a prime number p and a positive integer a such that p does not divide a. The remainder when a^{p-2} is divided by p is:

A. a
B. $p-1$
C. $a^{-1} \pmod{p}$
D. a^{p-1}

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Mathematical Logic

Which of the following is/are correct? ("$\times$" is the "multiply" relation and "/" is the "divides" relation)
S1: $[\{1, 2, 4, 8, 16, 32, 64, 128\}, \times]$ is a lattice.
S2: $[\{1, 3, 9, 27\}, |\ ]$ is a linear order.

A. S1 is correct, and S2 is correct.
B. S1 is correct, and S2 is incorrect.
C. S1 is incorrect, and S2 is correct.
D. S1 is incorrect, and S2 is incorrect.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Mathematical Logic

Which of the following is/are FALSE?

I. A relation that is symmetric, anti-symmetric, and transitive is called an equivalence relation.
II. When every two elements in a set are comparable, the relation is called a partial ordering.
III. The poset $(\mathbb{N}, \leq)$ is totally ordered, where $\mathbb{N}$ is the set of natural numbers.
IV. The subset relation $\subseteq$ is a total ordering on the power set of a set.

A. I and II only
B. I and III only
C. I, II and III only
D. I, II and IV only

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Discrete Mathematics

Let $(G, \cdot)$ be a cyclic group of order p^k, where p is a prime number and k is a positive integer. The number of subgroups of G is _____.

A. k
B. p
C. p^k
D. $k+1$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

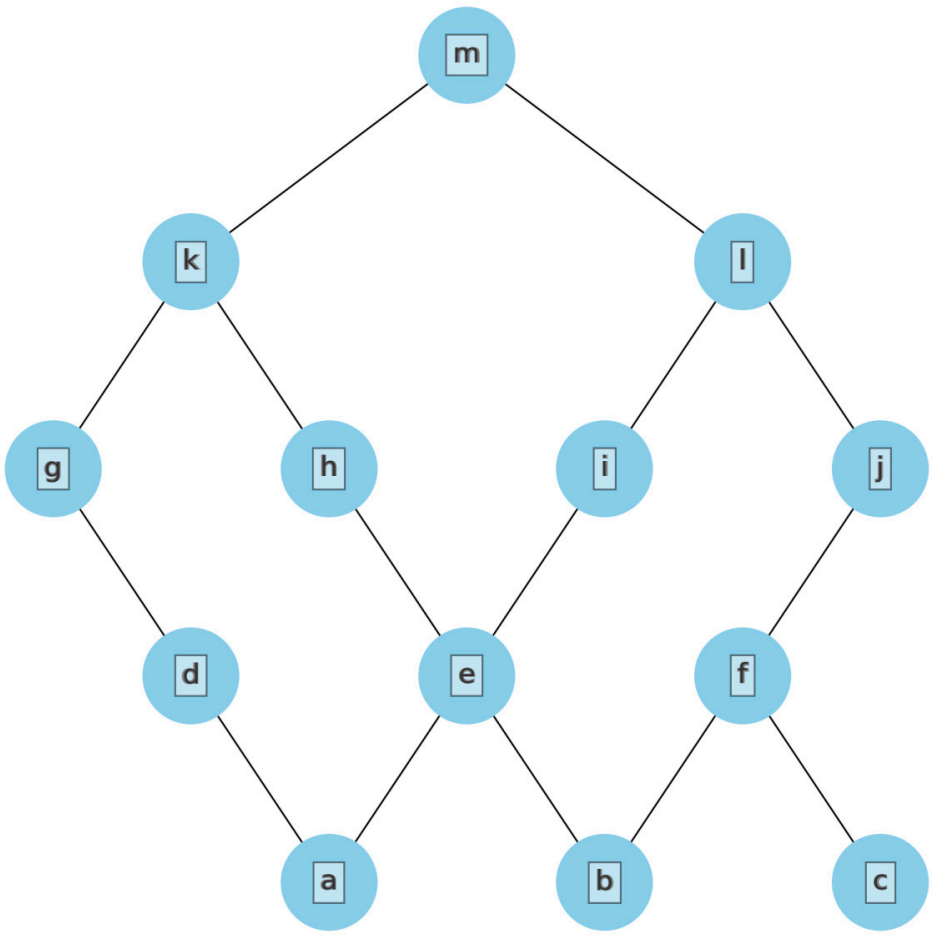
Q #11 | Numerical Type | Award: 2 | Penalty: 0 | Discrete Mathematics

The value of y_6 for the recurrence relation defined by $y_n = 4(y_{n-1}) + 2$ with the initial condition $y_1 = 2$ is _____

Your Answer: | Correct Answer: 2730:2730 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Mathematical Logic

The following is the Hasse diagram of the poset : $[\{a, b, c, d, e, f, g, h, i, j, k, l, m\}, \preceq]$



Given the diagram, the poset is:

A. Not a lattice.

B. A complemented and distributive lattice.

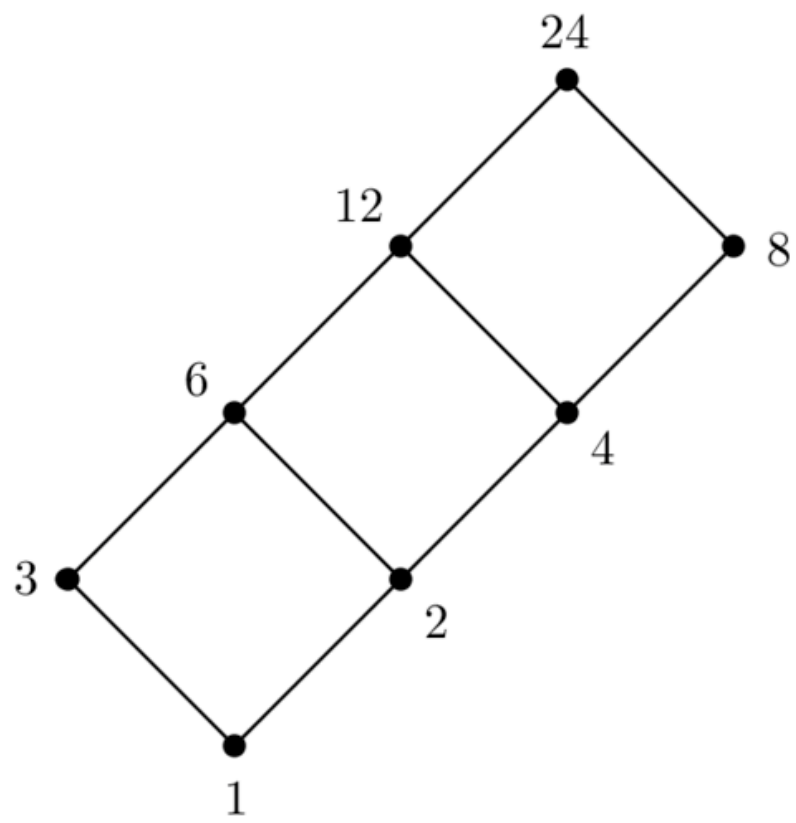
C. Neither a complemented lattice nor a distributive lattice.

D. A distributive lattice but not a Boolean algebra.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Numerical Type | Award: 2 | Penalty: 0 | Mathematical Logic

The following is the Hasse diagram of the poset $[\{1, 2, 3, 4, 6, 8, 12, 24\}, \ |]$, where $|$ is the divides relation.

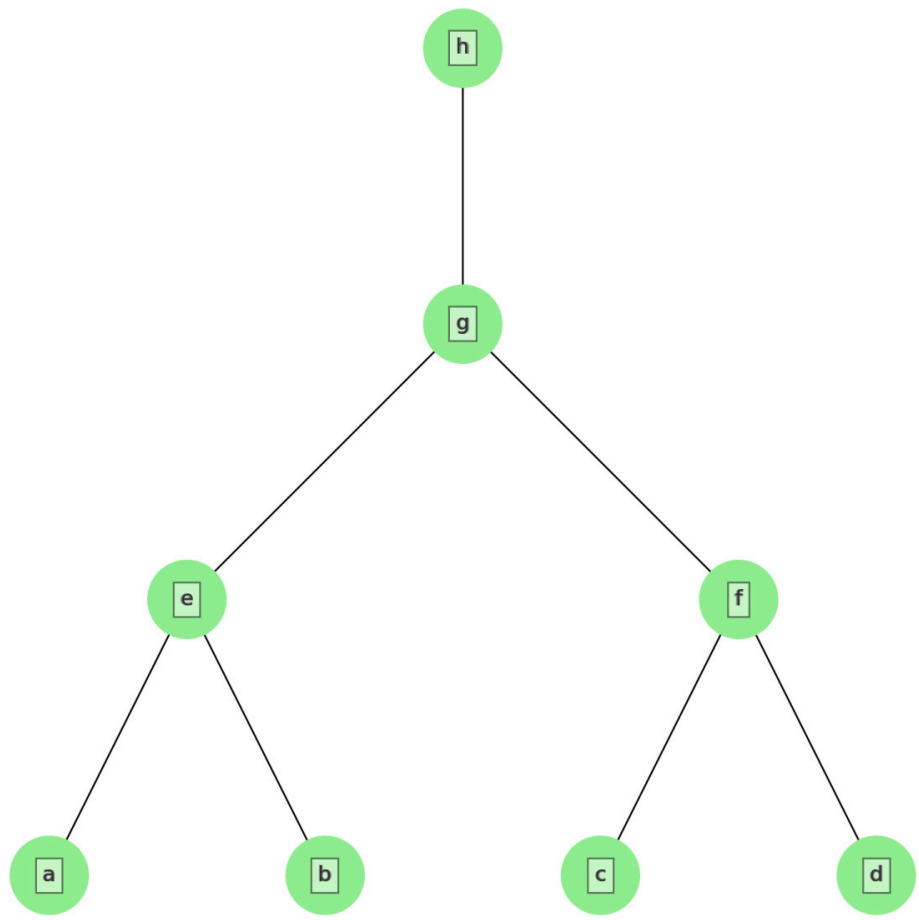


If the number of complements of $1, 2$ and 8 are α, β and γ respectively, then the value of $3\alpha^2 + 4\beta + 5\gamma$ is _____

Your Answer: | Correct Answer: 8:8 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Mathematical Logic

The following Hasse diagram describes a



A. Meet Semi lattice
B. Lattice
C. Join Semi lattice
D. None of the above

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

Which of the following is not correct regarding a complemented lattice?

A. $[P(\{1, 2, 3\}), \subseteq]$ is a complemented lattice.
B. $[\mathbb{Z}, \leq]$ is not a complemented lattice, where $\mathbb{Z}$ is the set of integers.
C. $[(0, 1), \leq]$ is a complemented lattice.
D. $[(\mathbb{N}, \leq)]$ is not a complemented lattice, where $\mathbb{N}$ is the set of natural numbers.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Mathematical Logic

The solution to the recurrence relation $b_n = b_{n-1} + 3^n$ with $b_0 = 2$ is _____ A. $b_n = 2 + 3^n$ B. $b_n = \frac{1}{2} + \frac{3^{n+1}}{2}$ C. $b_n = 2 \cdot 3^n$ D. $b_n = \frac{3^n - 1}{2} + 2$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 Multiple Select Type | Award: 2 | Penalty: 0 | Discrete Mathematics

Which of the following statements is NOT correct? (Mark all the appropriate choices)

A. Every cyclic group is finite.
B. The intersection of two subgroups of a group is also a subgroup.
C. Every subgroup of an abelian group is cyclic.
D. If a group G is cyclic, then G is abelian.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Mathematical Logic

The explicit formula for the sequence defined by the recurrence relation $x_n = 3x_{n-1} + 4x_{n-2}$ subject to the initial conditions $x_1 = 1$ and $x_2 = 2$ is:

A. $x_n = \frac{3}{20} \cdot 4^n - \frac{2}{5} \cdot (-1)^n$

B. $x_n = \frac{2}{5} \cdot 4^n - \frac{3}{20} \cdot (-1)^n$

C. $x_n = \frac{1}{10} \cdot 4^n + \frac{1}{4} \cdot (-1)^n$

D. $x_n = \frac{3}{5} \cdot 4^n + \frac{2}{5} \cdot (-1)^n$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 Numerical Type | Award: 2 | Penalty: 0 | Discrete Mathematics

Let G be a finite group on 15625 elements. The size of the smallest non-trivial proper subgroup of G is _____

Your Answer: | Correct Answer: 5:5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 Numerical Type | Award: 2 | Penalty: 0 | Discrete Mathematics

The remainder when 87^{100} is divided by 89 is _____

Your Answer: | Correct Answer: 2:2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 Numerical Type | Award: 2 | Penalty: 0 | Discrete Mathematics

How many isomorphic non-trivial finite abelian groups of order 100 are there?

Your Answer: | Correct Answer: 4:4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 Numerical Type | Award: 2 | Penalty: 0 | Discrete Mathematics

Let G be a cyclic group of order 240. How many distinct cyclic subgroups of prime order does G have?

Your Answer: | Correct Answer: 3:3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 Numerical Type | Award: 2 | Penalty: 0 | Discrete Mathematics

Let G be a finite cyclic group of order 60. The number of elements in G that generate the entire group G is _____

Your Answer: | Correct Answer: 16:16 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Discrete Mathematics

How many ways can we fill the following table so that the binary operation $\circ$ is associative?

$\circ$	**a**	**b**	**c**	**d**
a	a	b		d
b		c	d	
c	c		a	b
d		a		c

A. 2
B. 3
C. 1
D. 4

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 Numerical Type | Award: 2 | Penalty: 0 | Mathematical Logic

The remainder when 7^{100} is divided by 13 is _____.

Your Answer: | Correct Answer: 6:6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 Multiple Select Type | Award: 2 | Penalty: 0 | Discrete Mathematics

If $H = \{0, 1, 2, 3, 4, 5, 6\}$ is a group with respect to $\oplus 7$ (addition modulo 7), then which of the following is true? (Mark all the appropriate choices)

A. The inverse of 0 is 0

B. The inverse of 1 is 6

C. The inverse of 2 is 5

D. The inverse of 3 is 4

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Discrete Mathematics

Suppose N is a group that has exactly 72 distinct elements of order 6. How many distinct cyclic subgroups of order 6 does N have?

A. 10
B. 72
C. 6
D. 36

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Mathematical Logic

What is the solution to the recurrence relation $L_n = L_{n-1} + L_{n-2}$ with initial conditions $L_0 = 2$ and $L_1 = 1$?

A. $L_n = \frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^n + \frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n$

B. $L_n = \left(\frac{1-\sqrt{5}}{2}\right)^n - \left(\frac{1+\sqrt{5}}{2}\right)^n$

C. $L_n = \frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^n - \frac{-1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n$

D. $L_n = \left(\frac{1+\sqrt{5}}{2}\right)^n + \left(\frac{1-\sqrt{5}}{2}\right)^n$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Numerical Type | Award: 2 | Penalty: 0 | Discrete Mathematics

Given a sequence d_n defined by the recurrence relation $d_n = 2d_{n-1} - d_{n-2}$ with initial values $d_0 = 5$ and $d_1 = 2$, calculate the sum $d_{10} + d_9$.

Your Answer: | Correct Answer: -47:-47 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Discrete Mathematics

Consider the poset $P = \{\{1\}, \{1,2\}, \{1,2,3\}, \{2,3\}\}$ under the "subset" relation $\subseteq$. Determine the number of maximal and minimal elements in this poset.

A. Minimal: 1, Maximal: 1

B. Minimal: 2, Maximal: 1

C. Minimal: 2, Maximal: 2

D. Minimal: 1, Maximal: 2

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

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